NPTEL Online Certification Courses
Indian Institute of Technology Kharagpur

E-Business

Assignment- Week 10

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 2 = 20**

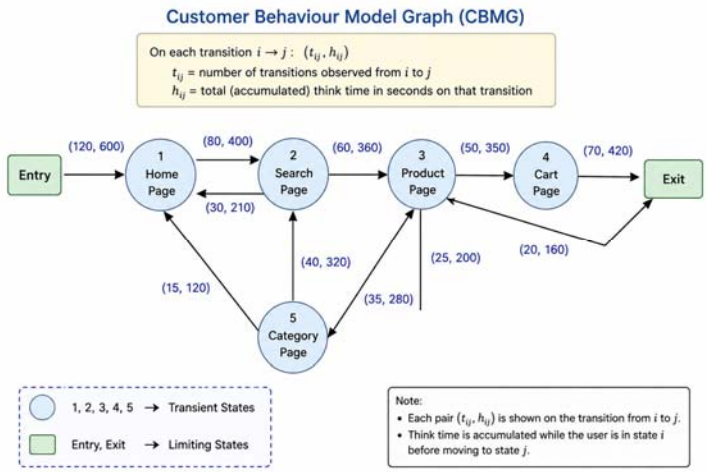
Question 1

A website administrator creates a Customer Behaviour Model Graph (CBMG) with Entry and Exit as special states. Which transition would violate the standard CBMG transition rules?

a) A transition from Exit back to Entry to represent a new session
b) A transition from Entry to a browsing state to begin navigation
c) **A transition directly from Entry to Exit without visiting any intermediate state**
d) A transition from a browsing state to Exit when a session terminates

Correct Option: c
Explanation: In a CBMG, no transition can be made directly from Entry to Exit. Users must first visit at least one intermediate state.
Week 10: Slide 4

Question 2



onsider the CBMG shown above. The transitions leaving the **Product Page (State 3)** are:

- Product Page → Cart Page: (50, 350)
- Product Page → Category Page: (35, 280)

- Product Page → Exit: (20, 160)

where each pair denotes **(transition count, accumulated think time in seconds)**.

What is the **expected time spent in the Product Page state before making a transition**?

a) 6.0 seconds
b) 6.5 seconds
c) 7.0 seconds
d) 7.5 seconds

Correct Option: d

Explanation:

The transition probabilities from Product Page are:

$$P(3 \to 4) = \frac{50}{105}, \quad P(3 \to 5) = \frac{35}{105}, \quad P(3 \to Exit) = \frac{20}{105}$$

The corresponding average holding times are:

$$350/50 = 7, \quad 280/35 = 8, \quad 160/20 = 8$$

Expected time spent in Product Page:

$$\frac{50}{105}(7) + \frac{35}{105}(8) + \frac{20}{105}(8) = 7.52 \text{ seconds}$$

Closest option = **7.5 seconds**.

Week 10: Slides 5, 7

Question 3

Assume each completed purchase in a website generates ₹2,500 average revenue. If 20,000 visitors arrive and the Buy-to-Visit Ratio is 4%, what is the expected revenue?

a) ₹1.5 crore
b) **₹20 lakh**
c) ₹50 lakh
d) ₹2 crore

Correct Option: b

Explanation:

Purchases =20000×0.04=800

Revenue = 800×2500=₹20,00,000

Week 10: Slides 12, 38

Question 4

In a CBMG-based customer segmentation study, suppose (V_b) represents browsing transitions and (V_s) represents searching transitions. A customer segment exhibits very high values of both. What does this most likely indicate?

a) Customers often terminate sessions immediately after login activities
b) Customers repeatedly revisit payment pages before confirming purchases
c) Customers spend most of their time on account-maintenance activities
d) **Customers engage extensively in exploration before making decisions**

Correct Option: d
Explanation: High browsing and searching activity indicates exploratory behaviour prior to purchase decisions.
Week 10: Slide 12

Question 5

Why are queueing network models generally preferred over single-queue models when analyzing large e-business infrastructures?

a) They eliminate the need to estimate service times for individual resources
b) They assume every server processes requests with identical characteristics
c) **They can represent multiple interacting resources and provide a more realistic performance model**
d) They guarantee optimal resource allocation without workload forecasting

Correct Option: c

Explanation: Queueing network models capture interactions among multiple resources such as web, application, and database servers, making them more realistic.

Week 10: Slide 31

Question 6

After forecasting future workload and estimating the expected arrival rate of requests, what should be the next step in the capacity planning process?

a) Increase hardware capacity immediately to accommodate future demand against quality-of-service requirements
b) Replace the existing forecasting model with a more complex alternative against quality-of-service requirements
c) Determine customer demographics to estimate future business growth against quality-of-service requirements
d) **Evaluate performance measures such as waiting time, utilization, and bottlenecks against quality-of-service requirements**

Correct Option: d

Explanation: After obtaining the expected workload, the performance model is used to evaluate waiting time, utilization, bottlenecks, and determine whether the quality-of-service requirements are satisfied.

Week 10: Slide 39

Question 7

An online retailer expects rapid growth in demand during the coming year. Which statement best captures the primary objective of capacity planning?

a) Determining the optimal website colour scheme for customer engagement
b) Forecasting future customer demographics for targeted advertising
c) **Predicting when workload may exceed available resources and planning accordingly**
d) Estimating future product prices under competitive market conditions

Correct Option: c
Explanation: Capacity planning predicts future saturation and determines how to delay it cost-effectively.
Week 10: Slide 15

Question 8

A site receives traffic from multiple business functions such as search, registration, and checkout. Why are resources often represented as queues in performance models?

a) To classify customers into homogeneous behavioural groups
b) To estimate future product demand under uncertain conditions
c) To identify relationships among item attributes in recommendation systems
d) **To analyze waiting times, utilizations, and response times systematically**

Correct Option: d
Explanation: Queueing models are used to estimate performance measures such as waiting time and utilization.
Week 10: Slides 23, 30–32

Question 9

A recommender system predicts how much a specific user will like an unseen movie based on previously expressed preferences. What type of recommendation decision is being made?

a) Top-M user identification for a newly introduced item
b) Community-wide ranking of the most frequently consumed items
c) Selection of items solely on the basis of demographic similarity
d) **Prediction of a user's preference for an unrated item**

Correct Option: d
Explanation: Predicting preference for an unseen item is a prediction-type recommendation decision.
Week 10: Slide 49

Question 10

A movie database initially contains **600 candidate features** extracted from descriptions. After applying TF-IDF-based feature selection, only the **top 80 features** are retained for learning the user profile.

What is the primary reason for retaining only these 80 features?

a) To ensure that every movie has exactly the same feature values.
b) **To remove less informative features and improve the efficiency of learning.**
c) To guarantee that the recommendation accuracy becomes 100%.
d) To make collaborative filtering independent of user ratings.

Correct Option: b

Explanation: Feature selection retains informative features while reducing computational complexity.

Week 10: Slides 61–62

**************END*********